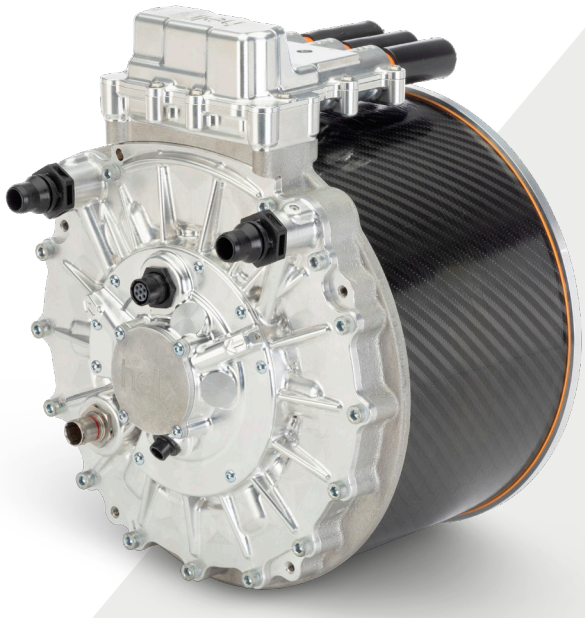
SPX242 Series

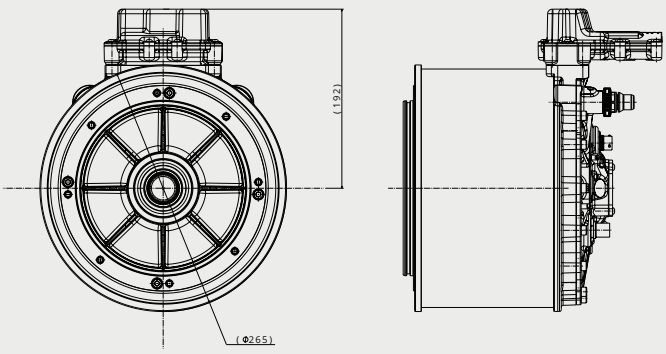
Scalable motors/generators

The SPX242 Series currently has three standard variants of differing lengths, each delivering world leading power and torque density while reliably providing high constant continuous power at high efficiency.

Leveraging Helix scalable core technology, the SPX242 series uses a combination of advanced electromagnetic design and a high-performance thermal management system, cooling both stator and rotor.

The SPX242 series of motors/generators is powerful, durable, and optimised for use in high-performance traction, propulsion, power generation and other specialist applications demanding high power, especially where space and weight are at a premium.





Key features

- Extremely compact, robust and scalable structure
- Simple installation and integration
- Peak Power of up to 500kW per unit for the longest derivative
- Up to 11.9kW/kg Peak & 8.5kW/kg Cont.
- Up to 17.9Nm/kg Peak & 11Nm/kg Cont.
- Peak efficiencies 97%
- Liquid cooled using water/glycol
- Low rotor inertia
- Compatible with various off the shelf transmissions
- Speed range of 0-17.75krpm
- Low mass, 24kg (SPC242-50)
- Manufactured in our volume facility

Powertrain Advantage

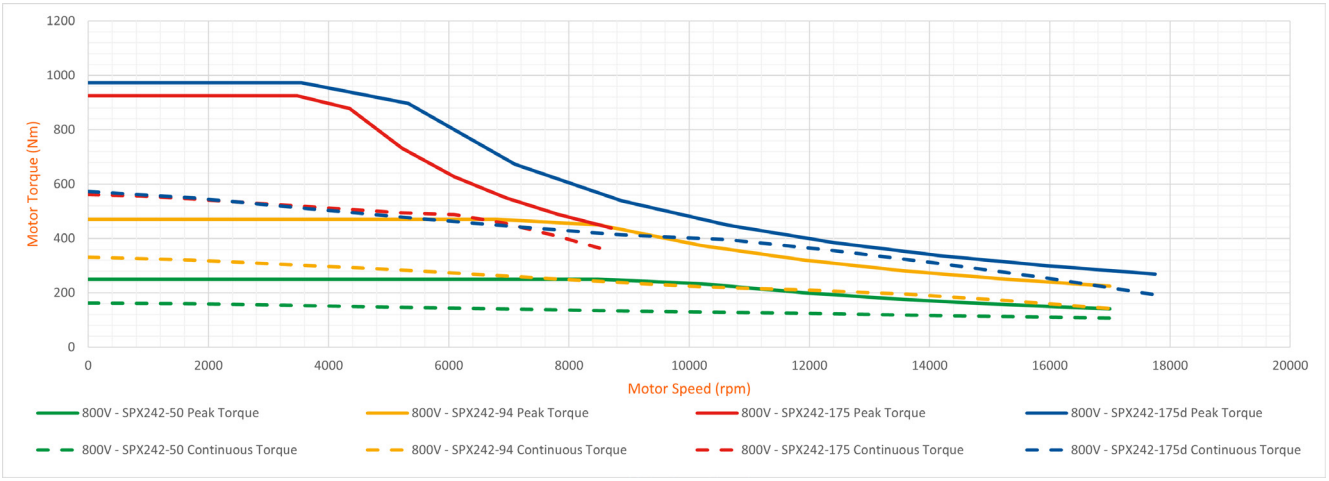
SPX242 Technical Data

Technical Data	Unit	SPX242-50 (800-600V)	SPX242-94 (800-600V)	SPX242-175 (800-600V)	SPX242-175D (800-600V)
Motor/Generator Type	-	3-Phase Radial Flux Permanent Magnet Synchronous Motor/Generator (SPX242-175D = 2x 3 phase)			
Applications	-	Motorsport, Automotive, Passenger Car, Commercial Vehicle (class 1-8), Off-Highway, Aerospace, Rail, Marine, Power Generation, Motorcycle, Hybrid, FCEV, REEV and BEV applications			
Physical Specifications					
Casing Diameter ø	mm	265	265	265	265
Package Length	mm	172	218	297	319
Mass[1]	kg	24	34	51	54
Performance Specifications					
Inverters Used: Also compatible with other Cascadia Motion/Sevcon/BorgWarner MCU's	-	750 ARMS Capable Inverter Incl. Helix SiC CTI-4 (10kg)			2x 750 ARMS Capable Inverter Incl. Helix SiC CTI-4
Maximum speed	rpm	17,000		8,700	17,750
Peak Power[3]	kW	250	400	400	500
Continuous Power[4]	kW	191	278	307	461
Peak Torque[3]	Nm	250	470	925	973
Continuous Torque[4]	Nm	163	331	556	548
Motor Only Power Density, Peak	kW/kg	10.6	11.9	7.8	9.2
Motor Only Torque Density, Peak	Nm/kg	10.6	14.0	18	17.9
Motor & MCU Power Density, Peak	kW/kg	7.4	9.2	6.5	6.7
Motor & MCU Torque Density, Peak	Nm/kg	7.4	10.8	15.1	13.1
Mechanical Specifications					
Axial/Radial Shaft Load	N	100 N axial | 200 N radial			
Mounting	-	Motor intended for horizontal installation mounted via drive-end housing (8x M6 at 205mm PCD)			
Drive Output	-	Internal involute female spline			
Ingress Protection (Installed)	IP	IP6K9K			
Motor Rotational Inertia	$kg.m^2$	0.0271	0.0458	0.0628	0.0628
Thermal Specifications					
Cooling Medium	-	50% Ethylene Glycol			
Coolant Inlet Temperature	°C	0 to +80			
Typical Coolant Flowrate	l/min	8-20			
Coolant Inlet Pressure	bar (gauge)	0.5-3.0			
Temperature sensor	-	PT100			
Minimum/Maximum ambient surrounding temperature	°C	-20 to +120			
Electrical Specifications					
Maximum DC Voltage (Motor)	VDC	925			
Maximum Phase Current (Motor)	Arms	657	773	762	1464
Peak Motor Efficiency	%	97			
Number of poles pairs	-	6			
Maximum Phase Cable diameter	mm^2	70			
Rotor Position Sensor	-	Tamagawa TS2225 Resolver (3 pole-pair)			
Line-Line Phase Voltage at Maximum Speed (at 20 °C)[2]	Vpk	743	1149	1193	1407
Storage Specifications					
Minimum/ Maximum Storage Temperature	°C	-20 to +60			

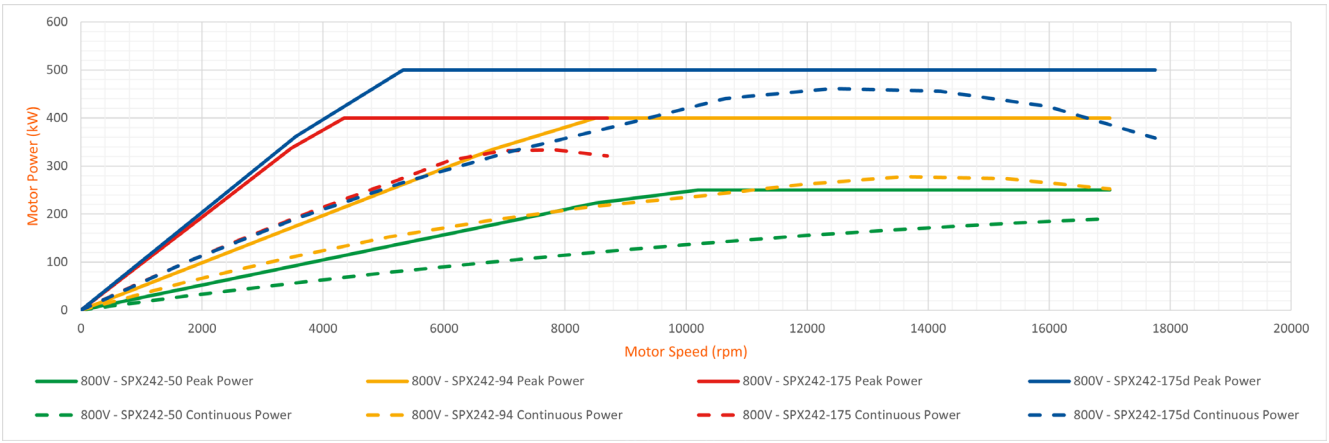
NOTES:
1. Dry Mass: Excludes phase cables, cable housing and coolant. Phase cables + cable housing are not part of the standard motor package.
2. Voltage measured between MGU terminals with magnet temperature at 20 °C
3. Peak Values (10 Seconds): Taken at 800VDC, 50°C coolant inlet
4. Continuous Values: Taken at 800 VDC, coolant flow rate 15 l/min at 50°C coolant inlet.

Not quite the specifications you need? No problem – the SPX platform leverages Helix's Scalable Core Technology architecture. This flexible approach offers a wide range of motor combinations to exactly meet your desired specifications (i.e. winding configuration, material grade and motor length) whilst minimising engineering investment required from our customers. Please contact us to learn more about our standardised configurations; our team will be glad to assist you in finding the motor suited to your exact requirements. For projects that demand mass production volumes, there is significant scope for reducing the bill of material cost and improving power and torque density in your application by using our Core Technology Bespoke approach. Please contact us to discuss how we can provide class-leading power and torque density motor/generators at a competitive cost for your volume application.

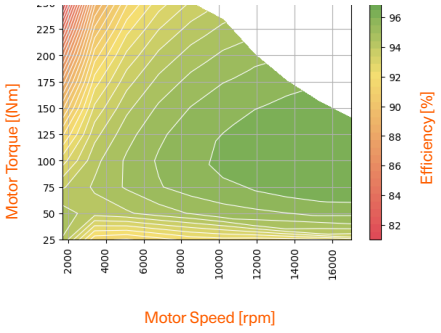
SPX242 Torque Curves



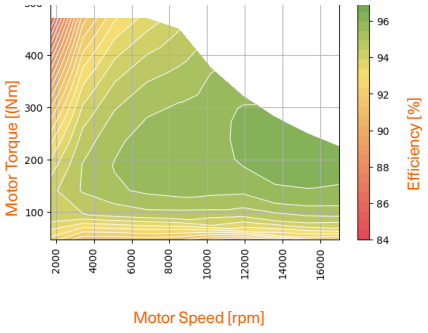
SPX242 Power Curves



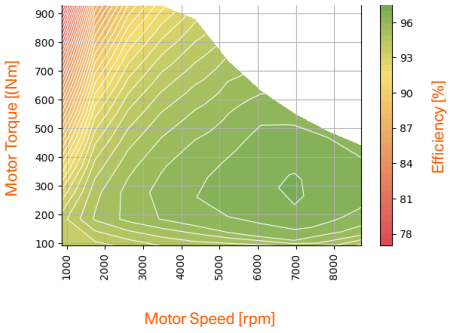
Efficiency Map SPX242-50 (800V)



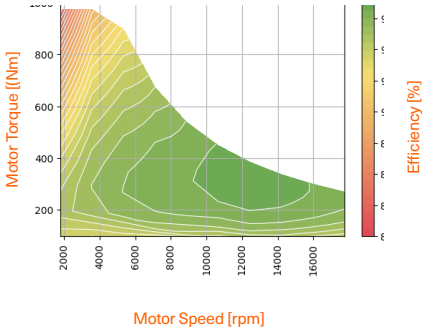
Efficiency Map SPX242-94 (800V)



Efficiency Map SPX242-175 (800V)



Efficiency Map SPX242-175D (800V)



Note: The performance ratings and graphs are based on typical operating conditions and typical test-validated values as noted (rather than minimum limits). There are several variables that may change the motor performance, including coolant flow rate, operating temperature, inverter settings and parameters, etc. For actual performance, the motor must be evaluated in its final system and application.

All data subject to change without notice.



CTI-4 Silicon Carbide Inverter

Our High Performance Inverter is suitable for hybrid and electric vehicles in automotive, motorsport and aerospace applications. The current platform of Silicon Carbide (SiC) MOSFET Inverters are a combination of technologies which have been developed and validated through R&D so that they can be applied with high confidence to new projects.

helix

Powertrain Advantage

+44 (0)1908 278600 contact@ehelix.com ehelix.com

Helix, Merlewood Drive, Shenley Wood, Milton Keynes MK5 6GR

SPX242/202601/EN